<u>Discrete Uniform distribution:</u>

The simplest of all discrete probability distributions is one where the random variable assumes its values with equal probabilities. Such a probability distribution is called rectangular or uniform distribution. It is called rectangular distribution since the histogram of this distribution is a set of rectangles of equal heights. It is uniform because the probability function of this distribution is constant or uniform over the entire range of the variable. Thus, if S is a sample space of a discrete random variable X, and $x_i \in S$ for $i = 1, 2, 3, \ldots, K$, that is $S = \{1, 2, 3, \ldots, K\}$, then $P(x_1) = P(x_2) = \cdots = P(x_K) = 1/K$ for a uniform distribution.

<u>Defn</u>:

Let the discrete random variable X assumes the values $x_1, x_2, \ldots, x_K$, each with equal probability $1/K$. Then the probability distribution of X, given by

$$f(x; K) = \begin{cases} 1/K ; & x = 1, 2, 3, \ldots, K \\ 0 ; & \text{otherwise} \end{cases}$$

is known as the uniform or rectangular distribution on the integers.

Great — let's go through your note step by step and make it very clear, with intuition.

Topic: Discrete Uniform Distribution

1. Introduction

- Among all discrete probability distributions, the **simplest** one is when a random variable takes its values with **equal probabilities**.
- Such a distribution is called a **rectangular distribution** or **uniform distribution**.

👉 Why "rectangular"?
Because if you plot its histogram, all bars (probabilities) have the same height — so it looks like equal-sized rectangles.

👉 Why "uniform"?
Because the probability is the same (constant) for each possible value.

---

2. Explanation

- Suppose the sample space is

$$S = \{1, 2, 3, \ldots, k\}$$

  where k values are possible.

- Then each outcome has the **same probability**:

$$P(X_1) = P(X_2) = \cdots = P(X_k) = \frac{1}{k}.$$

So, if there are k possible outcomes, each outcome has probability $1/k$.

---

3. Definition

Let X be a discrete random variable that can take the values

$$x_1, x_2, \ldots, x_k$$

with equal probability.

The probability distribution is:

$$f(x;k) = \begin{cases} \frac{1}{k}, & x = 1, 2, 3, \ldots, k \\ 0, & \text{otherwise} \end{cases}$$

This is known as the **discrete uniform distribution** (or rectangular distribution) on the integers.

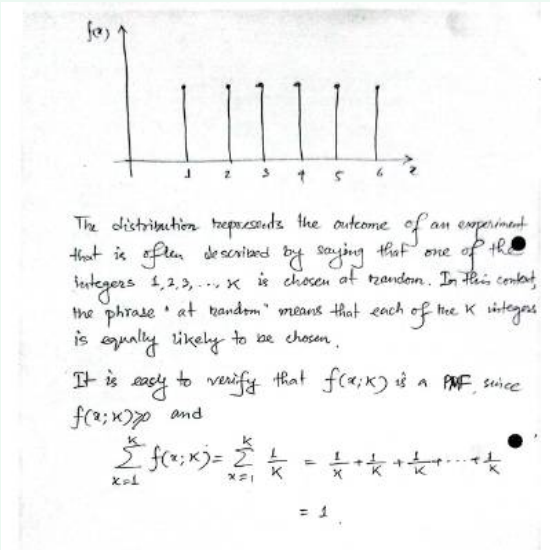
Intuition

Think of rolling a fair die 🎲:

- The outcomes are $\{1, 2, 3, 4, 5, 6\}$.
- Each has probability $1/6$.
- This is exactly a discrete uniform distribution with $k = 6$.

So whenever you have **equally likely outcomes**, you're dealing with a **uniform distribution**.

Would you like me to also explain the **mean and variance** of this distribution step by step (like in your other notes)?



The distribution represents the outcome of an experiment that is often described by saying that one of the integers 1, 2, 3, ..., K is chosen at random. In this context, the phrase "at random" means that each of the K integers is equally likely to be chosen.

It is easy to verify that $f(x;K)$ is a PMF, since $f(x;K) \geq 0$ and

$$\sum_{x=1}^{K} f(x;K) = \sum_{x=1}^{K} \frac{1}{K} = \frac{1}{K} + \frac{1}{K} + \frac{1}{K} + \cdots + \frac{1}{K}$$

$$= 1.$$

Perfect — let's carefully break this page down step by step with intuition.

---

1. Graph

At the top, we see a probability mass function (PMF) graph for a discrete uniform distribution.

- The x-axis: values $1, 2, 3, \ldots, 6$ (here $k = 6$, like a dice roll).
- The y-axis: probability, which is the same for each outcome:

$$f(x) = \frac{1}{6}.$$

So, the bars are all equal height — showing **equal likelihood**.

---

2. Description

The notes say:

- The distribution represents the outcome of an experiment where one of the integers $1, 2, 3, \ldots, k$ is chosen **at random**.

- The phrase “at random” means **each integer has equal probability** of being chosen.

👉 Example: Picking a card numbered 1 to 10 from a box without looking. Each card is equally likely.

---

3. Validity of PMF

To check if something is a valid probability distribution (PMF), two conditions must hold:

1. **Non-negativity**:

$$f(x;k) \geq 0$$

(probabilities can’t be negative).

2. **Total probability = 1**:
The sum of all probabilities must equal 1.
Here:

$$\sum_{x=1}^{k} f(x;k) = \sum_{x=1}^{k} \frac{1}{k}.$$

Expanding:

$$= \frac{1}{k} + \frac{1}{k} + \cdots + \frac{1}{k} \quad (k \text{ terms}).$$

$$= \frac{k}{k} = 1.$$

✅ So it’s a valid probability distribution.

Intuition

Think of it this way:

- If you divide a probability “cake” of size 1 into k equal slices, each slice is $\frac{1}{k}$.
- Adding up all k slices gives you back the whole cake (1).
- That’s exactly what this PMF check shows.

Would you like me to now go step by step into the **mean and variance** of the discrete uniform distribution (which usually comes right after this in notes)?

Problem:

Let X be a random variable representing the number on a ball drawn at random from a box containing 10 balls numbered 1 to 10. Find the probability distribution of X and hence $P(X<4)$.

Sol^n:

Clearly X assumes values $1, 2, 3, \ldots, 10$, each with equal probability $1/10$. Hence, the distribution is uniform with $K=10$. Thus,

$$f(x;10) = \frac{1}{10}; \quad x = 1, 2, 3, \ldots, 10$$

and $P(X<4) = \sum_{x=1}^{3} f(x;10) = \frac{1}{10} + \frac{1}{10} + \frac{1}{10} = \frac{3}{10}$

Ans

Ans

Problem:

A fair cubical dice is tossed once. Find the probability distribution of the number of points on the upper face when the dice comes to rest.

Sol^n:

The Sample space for the experiment is $S=\{1,2,3,4,5,6\}$ Each element of S occurs with equal probability $1/6$ and hence with $K=6$, the distribution is uniform and thus.

$$f(x;6) = 1/6 : \quad x=1,2,3,4,5,6.$$

Awesome — let's rewrite the two worked examples clearly, step-by-step, in the same style as your notes.

Problem 1

Statement.
Let X be the number on a ball drawn at random from a box containing 10 balls numbered 1 to 10. Find the probability distribution of X and hence $P(X < 4)$.

Solution.
Clearly X can take the values $1, 2, 3, \ldots, 10$.
Since the draw is "at random", each value is equally likely with probability $1/10$.
Thus the distribution is uniform with $k = 10$, and

$$f(x;10) = \begin{cases} \dfrac{1}{10}, & x = 1, 2, 3, \ldots, 10, \\ 0, & \text{otherwise.} \end{cases}$$

Now,

$$P(X < 4) = \sum_{x=1}^{3} f(x;10) = \frac{1}{10} + \frac{1}{10} + \frac{1}{10} = \frac{3}{10}.$$

$\boxed{P(X < 4) = 3/10}$

(Quick check: $\sum_{x=1}^{10} \frac{1}{10} = 1$, so it's a valid PMF.)

Problem 2

Statement.
A fair cubical die is tossed once. Find the probability distribution of the number of points on the upper face when the die comes to rest.

Solution.
The sample space is $S = \{1, 2, 3, 4, 5, 6\}$.
Each outcome occurs with equal probability $1/6$.
Hence, with $k = 6$, the distribution is uniform and

$$f(x;6) = \begin{cases} \dfrac{1}{6}, & x = 1, 2, 3, 4, 5, 6, \\ 0, & \text{otherwise.} \end{cases}$$

If you want, I can now derive the **mean and variance** for the discrete uniform $1, \ldots, k$ exactly like in your notes.

Continuous uniform distribution :

Uniform or rectangular distribution is one of the simplest distributions among the several parametric families of continuous distributions. It is particularly useful in theoretical statistics because it is easily amenable to further mathematical manipulation.

Def^n:

When the random variable X is continuous over the interval $[\alpha, \beta]$, then the random variable is said to have the uniform distribution if its density function assumes the form

$$f(x; \alpha, \beta) = \frac{1}{\beta - \alpha} ; \quad \alpha \leq x \leq \beta$$

where α and β are the two parameters of the distribut satisfying the condition $-\infty < \alpha < \beta < \infty$

Great — now this page is about the **Continuous Uniform Distribution**. Let's carefully explain it step by step with intuition.

---

1. Introduction

- The **Uniform (or Rectangular) Distribution** is one of the simplest continuous probability distributions.
- It belongs to the family of **parametric continuous distributions**.
- It's especially useful in **theoretical statistics** because it is simple and mathematically easy to work with.

👉 Intuition:
If you pick a number **completely at random** from an interval (say between 2 and 5), every point in that interval is equally likely. That is a **continuous uniform distribution**.

---

2. Definition

Let X be a continuous random variable defined over the interval $[\alpha, \beta]$.

Then, X is said to have a **uniform distribution** if its **probability density function (PDF)** is given by:

$$f(x;\alpha,\beta) = \begin{cases} \dfrac{1}{\beta-\alpha}, & \alpha \le x \le \beta \\ 0, & \text{otherwise} \end{cases}$$

- Here, α and β are the **parameters** of the distribution.
- The condition is: $-\infty < \alpha < \beta < \infty$.

---

3. Intuition

- The function $f(x) = \dfrac{1}{\beta-\alpha}$ is a **horizontal line**.
- This means probability is spread **evenly** across the whole interval.
- The total area under the curve (a rectangle of height $1/(\beta-\alpha)$ and width $\beta-\alpha$) is exactly 1, as required.

👉 Example:
If $X \sim U(2,5)$:

- Then the PDF is $f(x) = 1/(5-2) = 1/3$ for $2 \le x \le 5$.
- So the density is flat at height $1/3$ across the interval [2, 5].

Would you like me to continue with the **mean and variance** of the continuous uniform distribution (they usually come right after this in notes)?

The particular form of the distribution when X is continuous is often referred to as the uniform - distribution on the interval. Here X represents the outcome of an experiment that is often described by saying that a point is chosen at random from the interval (β, α).

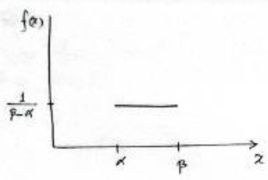


Fig.: Uniform density function

- Clearly, $f(x;\alpha,\beta)$ is a PDF since $f(x;\alpha,\beta) \geq 0$ and
$\int_{-\infty}^{\infty} f(x;\alpha,\beta)\, dx = \int_{\alpha}^{\beta} \frac{1}{\beta-\alpha}\, dx = \frac{\beta-\alpha}{\beta-\alpha} = 1$

Perfect — let's carefully go through this continuation on the **continuous uniform distribution**.

---

1. Explanation

- This is the particular form of the distribution when X is continuous on the interval $[\alpha, \beta]$.
- We often describe this situation by saying:
  "A point is chosen at random from the interval $[\alpha, \beta]$*."*
- Here, "at random" means every point in the interval is equally likely.

---

2. Graph

- The graph drawn shows a **rectangle**:
  - From $x = \alpha$ to $x = \beta$, the height is constant:

$$f(x) = \frac{1}{\beta - \alpha}.$$

  - Outside this interval, $f(x) = 0$.

So the density function is flat (a "uniform density function").

---

3. Verifying it is a PDF

To be a valid probability density function (PDF), two conditions must hold:

1. **Non-negativity:**

$$f(x; \alpha, \beta) \geq 0$$

   Clearly true since $\beta > \alpha$.

2. **Total area = 1:**

$$\int_{-\infty}^{\infty} f(x; \alpha, \beta)\, dx = \int_{\alpha}^{\beta} \frac{1}{\beta - \alpha}\, dx.$$

Now compute:

$$\int_{\alpha}^{\beta} \frac{1}{\beta - \alpha}\, dx = \frac{1}{\beta - \alpha} \times (\beta - \alpha) = 1.$$

✅ Verified — so this is a valid PDF.

Intuition

Think of it like spreading probability **evenly** across the whole interval $[\alpha, \beta]$.
The rectangle has **width =** $\beta - \alpha$ and **height =** $1/(\beta - \alpha)$, so its area is exactly 1.

That's why it's called **uniform** — no value inside the interval is more likely than another.

Would you like me to now continue with the **mean and variance derivation** for the continuous uniform distribution (usually comes next in notes)?